

FORMAL TECHNICAL STATEMENT AND ACCIDENT RECONSTRUCTION ANALYSIS

Incident Location: Interchange of State Highway 121 & Bass Pro Drive / W. Bethel Road, Grapevine, Texas **Date & Time of Incident:** August 28, 2026, at 05:00:15 AM CDT (GPS Telemetry Verified)

Vehicle Trajectory: Eastbound on Bass Pro Drive across the SH-121 overpass, executing a protected left turn onto the Northbound SH-121 frontage road / entrance ramp **Intended Trip**

Routing: En route to pick up passenger at Aloft Dallas DFW Airport Grapevine (N. State Highway 121) for transport to DFW International Airport **Uber Commercial Status:** Period 2 / Tier 2 (Active Dispatch). Upfront fare: \$12.07. Rider cancellation recorded post-impact at 05:05:00 AM

1. High-Resolution GPS Telematics Breakdown

Data recorded via 1 Hz GPS telemetry (20260828.dat parsed to gps-20260828.csv, 13,285 continuous records) establishes the second-by-second physical dynamics of the vehicle prior to impact:

04:58:40 – 04:59:10 AM (Eastbound Approach): Vehicle traveling eastbound on Bass Pro Drive at speeds between 23.0 mph and 40.7 mph on a heading of 92° ESE.

04:59:18 – 04:59:43 AM (Preceding Signal Stop): Full 25-second stop at coordinates 32.955082, -97.041977 (362 meters west of impact). This stop occurred at the traffic signal on the west side of the overpass (Southbound SH-121 frontage road).

04:59:44 – 05:00:05 AM (Overpass Bridge Crossing): Departed the west signal on green, accelerating from 3.0 mph to 39.1 mph due east (90° E) across the SH-121 overpass bridge deck (inter-intersection storage zone).

05:00:05 – 05:00:14 AM (Controlled 9-Second Deceleration): Continuous, linear deceleration from 39.1 mph down to 17.2 mph while approaching the dual left-turn bay.

05:00:13 – 05:00:14 AM (Left-Turn Execution): Vehicle heading swung left from 84° to 70° E as the vehicle initiated entry into the Northbound SH-121 on-ramp turn lane at 17.2 mph.

05:00:15 AM (Point of Impact): Broadside impact occurred at an calculated speed of 15.0–17.2 mph. Vehicle was brought to an immediate, permanent halt at a heading of 22° NNE (rotated 68° into the left turn). Vehicle telematics confirm zero movement following 05:00:15 AM.

2. Interchange Architecture & Signal Coordination Logic

The intersection at SH-121 and Bass Pro Drive / W. Bethel Road operates as an interconnected **TxDOT Diamond Interchange**:

Integrated Controller Network: The traffic signals governing the west side of the overpass (Southbound 121) and the east side (Northbound 121) are electronically linked and governed by a master NEMA TS2 traffic controller cabinet.

Progression Call Triggered by West-Side Stop: The 25-second full stop at 04:59:18 AM held the vehicle at the west entrance signal. When that signal turned green at 04:59:44 AM, the vehicle crossed advance in-pavement inductive loops on the bridge deck.

Downstream Protected Green Arrow Activation: The interconnected controller calculated vehicle speed (39.1 mph) and arrival timing across the 362-meter bridge span. This automated logic called the **downstream protected green arrow** for eastbound left turns onto NB-121, energizing the turn phase precisely as the vehicle decelerated to 17.2 mph between 05:00:05 AM and 05:00:14 AM.

3. Phase Isolation & Physical Impossibility of Opposing "Yellow Light"

Under Texas Department of Transportation (TxDOT) signal engineering standards:

Exclusive Protected Right-of-Way: The dual left-turn lane onto NB-121 operates on a **Protected-Only** phase. Activation of the green turn arrow isolates the movement from all conflicting traffic.

Mandatory Solid Red Signal: While the eastbound protected left-turn green arrow is active, opposing westbound straight-through traffic on W. Bethel Road is assigned a **Solid Red Light** by the controller.

Malfunction Management Unit (MMU) Hardware Interlock: The physical MMU inside the cabinet continuously monitors signal line voltages. It is physically and electronically impossible for the controller to output a yellow or green signal to opposing straight traffic while the left-turn green arrow is energized. Any such voltage conflict would instantly trip the cabinet into emergency 4-way flashing red mode.

Reconstruction Finding: The opposing driver's claim to law enforcement that her signal was "yellow" represents a physical and technical impossibility under the established diamond interchange timing sequence.

4. Commercial Status Verification & App UI Artifact Explanation

Official Uber trip logs and device screenshots confirm active commercial dispatch status (Period 2 / Tier 2) at the time of impact:

Dispatched Route: Driver accepted an active dispatch to pick up a passenger at **Aloft Dallas DFW Airport Grapevine** (N. State Highway 121) for transport to **DFW International Airport**.

Impact vs. Cancellation Timestamps: Telemetry fixes impact at **05:00:15 AM**. The rider cancellation occurred post-impact at **05:05:00 AM** (4 minutes, 45 seconds *after* the collision).

Uber App Receipt Artifact: The Uber trip receipt lists identical origin and destination fields (N State Highway 121, Coppell, TX) and a \$0.00 fare. Under Uber driver app architecture, when a trip is canceled prior to passenger pickup, the system automatically wipes the intended destination from the driver receipt and populates both location fields with the GPS coordinates where the cancellation event was logged by the server.

Coverage Application: Because the crash occurred while actively navigating to an accepted rider pickup, the vehicle was strictly operating under **Period 2 (Tier 2)** commercial status, establishing primary commercial liability and coverage obligations.

5. Summary & Technical Conclusion

Objective 1 Hz GPS telematics demonstrate a controlled, 9-second deceleration into a protected left turn at 17.2 mph, following a coordinated green progression across the SH-121 overpass. Under Texas diamond interchange signal architecture, the opposing straight-through movement was held at a **solid red light**. The physical telemetry, signal timing logic, and trip dispatch logs confirm zero fault on the turning driver and establish complete liability on the opposing vehicle for failing to yield to a red signal.

6. Appendix: Formatted 1 Hz GPS Telemetry Data Log

Data Source: D:\GPS\20260828.dat parsed to **gps-20260828.csv** (13,285 records, 1 Hz sample rate). **Speed Conversion Verification:** 1 knot = 1.15078 mph. Verified against position deltas (0.000130° latitude in 1 s = 14.47 m/s = 28.1 kn vs. logged 27.874 kn).

| Time (CDT) | Speed (knots) | Speed (mph) | Heading | Telemetry Event / Physical Location | | | | | |

04:58:40 – 04:59:10 | 20.0 \rightarrow 35.4 kn | 23.0 \rightarrow 40.7 mph | 92° ESE | Traveling eastbound on Bass Pro Drive. | | **04:59:18 – 04:59:43 | 0.0 kn | 0.0 mph (25 s) | — | Full stop at 32.955082, -97.041977** (362 m west of impact; SB-121 frontage road signal). | | 04:59:44 – 05:00:05 | 2.6 \rightarrow 34.0 kn | 3.0 \rightarrow 39.1 mph | 90° E | Departing west signal on green; accelerating across overpass bridge deck. | | **05:00:05 – 05:00:14 | 34.0 \rightarrow 15.0 kn | 39.1 \rightarrow 17.2 mph | 90° \rightarrow 70° E | Nine (9) seconds of continuous linear braking approaching dual left-turn bay.** | | 05:00:13 | 15.6 kn | 18.0 mph | 84° E | Heading begins swinging left into turn lane. | | 05:00:14 | 15.0 kn | 17.2 mph | 70° E | Left turn in progress under protected signal. | | **05:00:15 | 0.0 kn | 0.0 mph | 22° NNE | IMPACT. Vehicle brought to permanent stop at 22° NNE. Telemetry terminates movement.** |